

FLAGELLA-CASCADE INVESTIGATION · PRELIM AIM 2

Why the flagella cascade ran away, and what we changed to fix it

The biology, the whole-cell-model (WCM) limitations we hit, and the replacements we built — walked through for discussion with your PI.

Maya Abdalla · v2ecoli · 2026-08-06/07

The flagella cascade had no brake

Flagellar gene expression in *E. coli* is a real transcriptional cascade — a master regulator turns on structural genes, which eventually turns on more genes. In the model, as originally built, nothing ever turned it back off.



First instinct: cap flagella count with a hard-coded ceiling. **Rejected** — not a real mechanism, just a patch. The right fix had to be a real, citable biological brake.

The WCM's standard tool: Gillespie complexation

Most protein complexes in the whole-cell model (ribosomes, transcription factors, receptor complexes — hundreds of them) are built by one shared, generic process: `ecoli-complexation`.

HOW IT WORKS

Each tick, it runs the **Gillespie stochastic simulation algorithm (SSA)** over every complexation reaction — computing the probability ("propensity") that each reaction fires, given current subunit counts, then sampling which ones actually do.

WHY IT NORMALLY WORKS FINE

Most real complexes need small numbers of each subunit — 2, 4, a dozen copies. The propensity math handles that scale without any trouble.

This is the process the flagellum's assembly reactions were originally routed through too — and it's where things broke.

Real flagellar stoichiometry breaks Gillespie's math

The vendored (default) flagellar reactions used a lazy placeholder — **every subunit at -1 copy** — because nobody had filled in the real numbers. Once we corrected that to real biology, ParCa started failing with:

```
"failed simulation: total propensity is NaN"
```

Why: Gillespie's propensity for a multi-subunit reaction is a *product* of combinatorial terms, one per reactant. A single coefficient of 60 is fine on its own (already existed safely elsewhere in the model) — but **several large coefficients multiplied together**, or one truly huge one (FliC's real count, ~20,000 per filament), overflows the calculation. Confirmed by direct testing, not guesswork.

This is a numerical limitation of the generic mechanism — not a sign the biology is wrong. Most of the other flagellar/chemotaxis complexation reactions (FlhD4C2 itself, the four chemotaxis receptors, CheZ) have small coefficients and never needed to move.

Correcting the stoichiometry to real, cited values

REACTION	COMPONENT	WAS	REAL VALUE	SOURCE
CPLX0-7452 (flagellum)	FliC (filament)	-1	-20,000	PMC7696725
	FlgK / FlgL (junction)	-1	-11 each	cryo-EM
FLAGELLAR-MOTOR-COMPLEX	FliF (MS-ring)	-1	-34	cryo-EM, C34 symmetry
	FlgH / FlgI (rings)	-1	-26 each	cryo-EM, C26 symmetry
	MotA / MotB (stator)	-1	-55 / -22	derived, medium confidence
CPLX0-7450 (new stage)	FliG / FliM / FliN	didn't exist	-34 / -34 / -111	PMC10128058

Also fixed 2 structural gaps: the export apparatus was built but never consumed (orphaned); the motor-switch complex (C-ring) didn't exist as its own stage at all.

Route the 3 broken reactions around Gillespie entirely

Real stoichiometry stays (slide 5) — but these 3 reactions get computed **deterministically** by dedicated Steps instead of stochastically:

REPLACES	NEW STEP	KIND OF PROCESS
CPLX0-7450_RXN	flagella_motor_switch_assembly.py	instant conversion, as much as material allows
FLAGELLAR-MOTOR-COMPLEX_RXN	flagella_motor_complex_assembly.py	instant conversion, as much as material allows
CPLX0-7452_RXN (filament)	flagella_filament_nucleation.py	rare, single discrete event (~1/10min)
	flagella_filament_elongation.py	continuous, rate-limited growth over time

Filament growth is the important exception. A complexation reaction is instantaneous — all subunits at once or nothing. Real filament growth isn't instantaneous; it needed a genuinely different kind of mechanism, not just a numerical workaround. That's next.

Filaments grow by injection-diffusion — and slow down as they grow

Renault et al. 2017 (*eLife* 6:e23136) measured how real flagellar filaments actually grow: flagellin subunits are pumped through the hollow filament core to the tip, and have to **diffuse the whole length already built** to get there.

$$dL/dt = a / (b + L)$$

$a \approx 26,450 \text{ subunit}^2/\text{s}$ · $b \approx 575 \text{ subunits}$

FILAMENT LENGTH	GROWTH RATE
0 (just started)	46.0 subunits/s
5,000	4.7 subunits/s
15,000	1.7 subunits/s
19,000 (near done)	1.3 subunits/s

The slowdown is **real biology**, not a modeling artifact — the longer the tube, the harder the diffusion.

DIRECT ANSWER TO: "IS 20,000 FLIC SUBUNITS PER FLAGELLUM FEASIBLE?"

Yes — but a full filament genuinely takes over 2 hours, even best-case

Solved Renault et al.'s own rate law directly for how long ONE filament takes to reach 20,000 subunits, assuming **unlimited** FliC supply (no synthesis bottleneck at all):

133.3 min

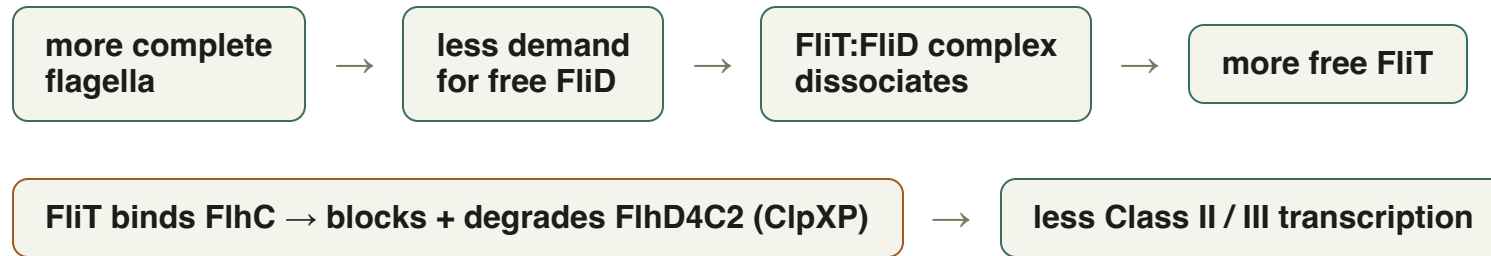
($\approx$ 2.22 hours) — the theoretical *minimum*, best case, from the growth law alone

Our 2-hour (120 min) multi-generation test window was shorter than this minimum. Zero completions in that run isn't evidence the stoichiometry is unrealistic — it's expected, because even a perfectly-supplied cell can't finish a filament that fast. The real, still-open synthesis-rate gap (slide 12) stretches real completion time out even further beyond this floor.

Recommendation: don't reduce the 20,000 coefficient — it's real, cited structural biology (PMC7696725), not a guess. Extend test windows instead (3–4+ hours) to actually observe completions.

FliT: an export chaperone that moonlights as a brake

FliT's day job is escorting FliD (the filament cap) out to the growing tip. But it also directly regulates FlhD4C2 — and that link is what closes the loop.



FliT is **catalytic and recycled**, not consumed — matches the real biochemistry (0.02 μ M FliT sufficient to degrade a 100× molar excess of FlhC).

HOW IT ACTUALLY RUNS, EACH SIMULATION TICK

One consistent, ordered chain — not a race condition



Each step in this chain sees the *already-updated* result of the one before it, same tick — the ordering is enforced explicitly, not assumed.

The lag that matters

New mRNA written this tick still needs several more ticks of elongation, then translation, then complex formation, before it becomes new FliT/FliD protein. The checkpoint always reads the **accumulated standing pool from prior ticks**, never anything made in the same tick — matching the real biological delay between transcription and a finished protein.

Two division-boundary bugs, found and fixed

BUG 1 — FEATURE DROPPED AT DIVISION

Daughter cells silently reverted to the plain model at every division, then crashed outright (stale step rebuilt with placeholder gene IDs). Fixed by threading the feature flag through the daughter-rebuild call, same pattern already used elsewhere for swapped processes.

BUG 2 — FLAGELLA DUPLICATED, NOT SPLIT

In-progress flagella had no divider registered — fell through to a generic "copy to both daughters" rule. Physically impossible, and would have exponentially inflated the count across generations. Fixed with a real binomial 50/50 split per flagellum.

Both found *before* being asked, by tracing through what a real multi-generation test would need — either one alone would have silently invalidated every multi-gen result.

A REAL CALIBRATION GAP, FOUND AND PARTIALLY CLOSED

FliC's transcription was capped well below real demand

MEASURED BEFORE

0.71 /s

real-time, sustained measurement

MEASURED AFTER 10× FIX

6.0 /s

≈8.5× real improvement

TARGET (RENAULT-IMPLIED)

~46 /s

still ≈7.6× short

Root cause: the Class III override caps out at fliC's normal, non-motile-induced expression level — it can never exceed it. **Why we stopped at 10×:** all mRNA genes share one fixed transcription budget; 10× already gives fliC ≈2.3% of the *entire cell's* mRNA output. The full 65× needed to close the gap would take ≈13.5% — judged unrealistic, would starve hundreds of other genes.

2-hour, 4-cell test: mechanics work, timing doesn't add up yet

- Population grew 1→4 cells over 3 real divisions — both division fixes held up under real load.
- Nucleation kept firing: 30 flagella under construction by the end.
- **Zero completed.** Longest filament reached 10,730 / 20,000 subunits.

Given slide 8's finding — a filament needs ≥ 133 min *even with unlimited FliC*, and real supply is still $\approx 7.6\times$ short of peak demand — **zero completions in a 120-minute window is roughly what the math predicts**, not a surprise result.

Not yet known: whether completions ever catch up to division-driven dilution over a longer run, or a properly-scaled test window.

Status & open questions

Done, verified

- Real, cited stoichiometry throughout the assembly pathway
- Real FliT:FliD negative-feedback checkpoint replacing the arbitrary cap
- Physical nucleation + Renault-kinetics elongation, out of Gillespie for numerical reasons
- Both division-boundary bugs fixed and verified under real multi-gen load
- FliC synthesis gap: real, measured, closed from $\sim 65\times$ to $\sim 7.6\times$

Still open

- FliG copy number: 34 (2 lit. sources) vs. 26 in Maya's spreadsheet — unresolved
- Whether flagella count actually self-limits, once run for a properly-scaled time window
- Whether the FlgM $\rightarrow$ FliA $\rightarrow$ Class III loop has its own brake, now that FliC supply is the binding constraint